NICHOLAS CARRERA

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EDUCATION

DePaul University | B.S. UX Design, Minor in CS | GPA: 3.835, Magna Cum Laude — Chicago, IL | June 2026

PROJECTS AND EXPERIENCE

Freelance Design Engineer | Self-employed — Jun 2026 - Present

- Built the online magazine substanzmagazine.com by hand in HTML, CSS, and JavaScript, and scored its teaser video
- For ANKLE, a live art and culture platform: reviewed its public API, sent the founder a ranked list of engineering fixes, and built an MCP bridge that lets Claude work inside the platform
- For Scene Portal, a pre-launch film pre-production tool: re-ran the founder's 3D scan (Gaussian splat) speed test on a second machine and confirmed that screen resolution, not file size, drives render speed

LLM Semantic 3D Geometry | DePaul URAP research assistant, solo, advised by Nathan Koch — Sep 2025 - Mar 2026

- Built a FastAPI server in Python that switches between local models (Ollama) and the Claude API with no code changes, so each experiment changed one thing
- Compared Mistral 7B with Claude Haiku, Sonnet, and Opus on one renderer and found my own 7-number output format, not the model, limited the shapes
- Moved rendering to TouchDesigner, then rebuilt it as a public React Three Fiber demo at urap-geometry.pages.dev that anyone can try

MADMADHOUSE Archive and Booking Site | madmadhouse.com — Sep 2026 - Present

- Designed and built a flyer archive and booking page for my DJ collective using React, TypeScript, and Vite, so promoters can see 44 nights across 21 venues in one place
- Wrote a Python validator that checks every Instagram post made it onto the site, and proved it catches 8 of 8 planted faults, so no flyer silently drops off
- Stopped the page from jumping as it loads (layout shift 0.113 to 0) and set every text color to pass WCAG AA contrast, so the archive stays readable on phones

CAT SSO and Access Control | DePaul x Caterpillar capstone, team of four — Jan 2026 - Jun 2026

- Led my team's reframe of in-cab sign-in after a dealer instructor reported under 5% of operators used it, designing around why an operator would sign in at all
- Helped design a three-layer access system (who you are, what you can do, how long you stay signed in) and a 10-flow Figma prototype with offline sign-in
- Presented our design to Caterpillar UX, then redesigned the status bar and the in-cab NFC reader symbol after their systems expert's review

Wabash Lights Data Visualization | DePaul, public installation, solo — Mar 2025 - Jun 2025

- Cleaned Urban Displacement Project housing data for 1,982 Chicago census tracts (1990 to 2018) in Excel and Tableau, then coded a p5.js light animation driven by it for four LED rails, approved by faculty to run live under the CTA tracks

SKILLS

Python, TypeScript, React, React Three Fiber, FastAPI, Claude API, TouchDesigner, p5.js

CERTIFICATIONS

AI Fluency: Framework and Foundations (Anthropic), Claude 101 (Anthropic), Certified SolidWorks Associate, Mechanical Design (Dassault Systemes, 2020)